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Exp No: 3	**Experiments on CISCO PACKET TRACER - Hub and Switch**

AIM:

To understand the environment of Cisco Packet Tracer, to design simple networks using a hub and a switch with 4 to 6 hosts, and to analyse the behaviour of these devices by tracing the communication in simulation mode.

ALGORITHM / PROCEDURE:

1. Open Cisco Packet Tracer. Study the workspace: device selection box (bottom left), logical workspace (centre), Realtime / Simulation switch (bottom right).
2. Select **End Devices** and drag 5 PCs and a Hub on to the workspace.
3. Select **Connections** and use the **Automatically Choose Connection Type** (copper straight-through) to connect each PC to the hub.
4. Click a PC, go to **Desktop > IP Configuration** and set the IP address `192.168.1.1` to `192.168.1.5` with subnet mask `255.255.255.0`.
5. Open **Desktop > Command Prompt** of PC0 and ping the other PCs to check the connectivity.
6. Switch to **Simulation** mode, click **Add Simple PDU** (envelope) and click PC0 then PC3. Press **Capture / Forward** and watch how the packet moves.
7. Repeat steps 2 to 6 by replacing the hub with a **Switch** (Experiment 2).
8. Build a network that has both a hub and a switch (Experiment 3), connect hub and switch with a crossover cable, send a PDU from a hub PC to a switch PC and compare the behaviour.
9. Observe the event list and the MAC address table of the switch `show mac-address-table`.

DIAGRAM:

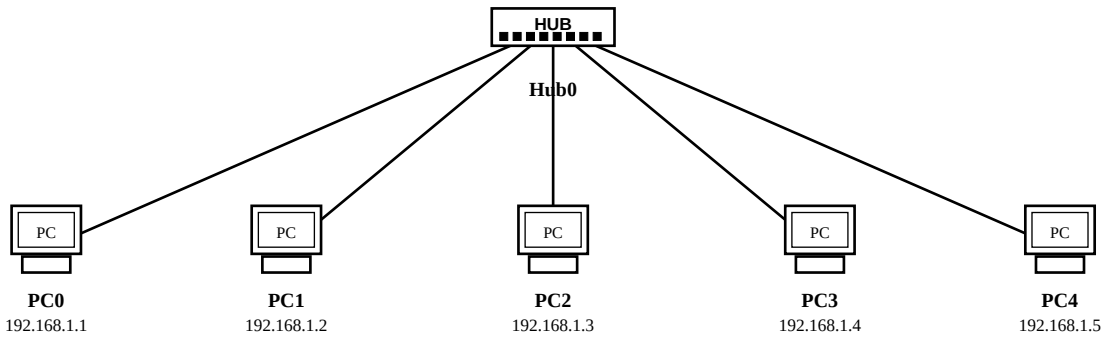

Fig 1: Network using only a HUB (5 hosts)

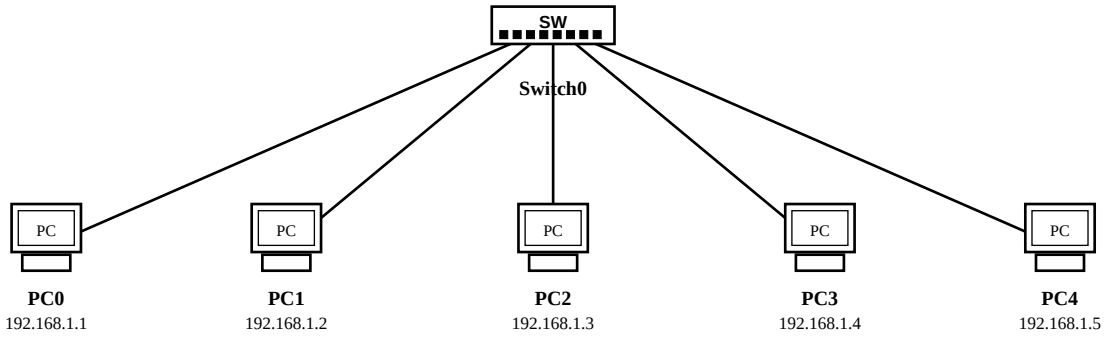

Fig 2: Network using only a SWITCH (5 hosts)

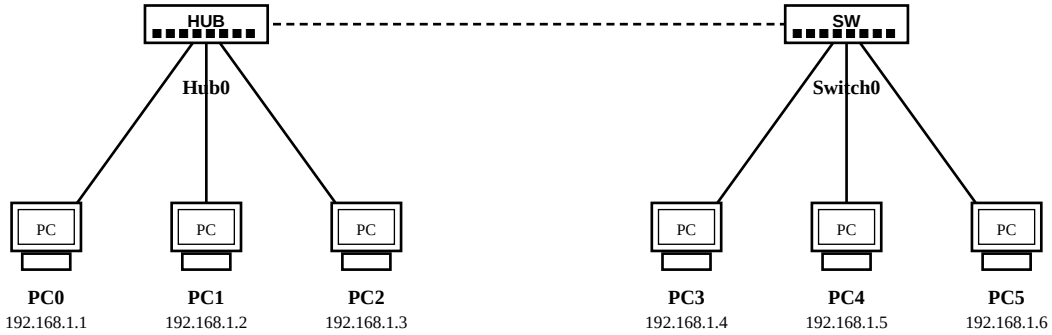

Fig 3: Network using both HUB and SWITCH (6 hosts)

Network topology implemented in our college

The college network follows an **extended star (hierarchical) topology**. The computers of each lab are connected to a lab switch with straight-through UTP cables (star). All lab switches are connected to the core switch kept in the server room and the core switch goes to the router / firewall and then to the ISP.

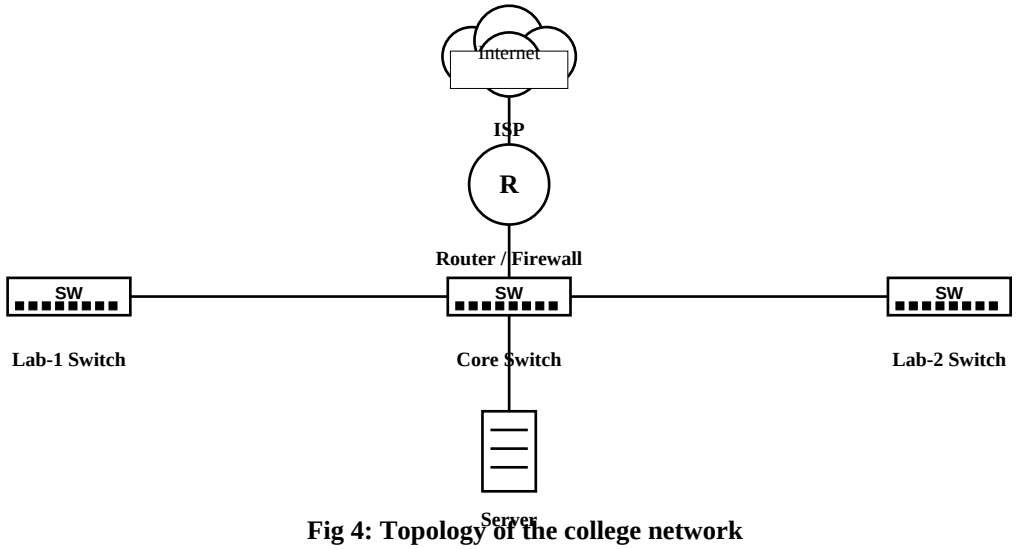

Fig 4: Topology of the college network

OUTPUT:

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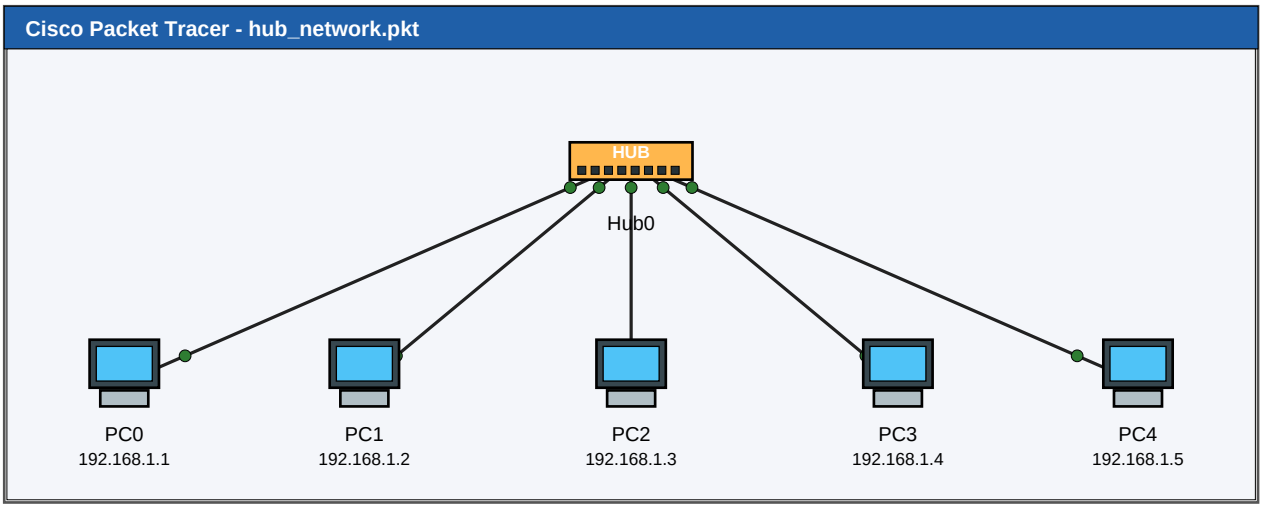

Experiment 1 : hub network in Packet Tracer (all link lights green)

PC0 - Command Prompt

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.4: bytes=32 time=6ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 2ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The first ping loses one packet because the ARP request has to be resolved first. The second ping shows 0% loss.

Simulation mode - Hub (PC0 to PC3)

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Hub0	ARP		0.000	N	0
	Successful	Hub0	PC1	ARP		0.001	N	1
	Successful	Hub0	PC2	ARP		0.001	N	2
	Successful	Hub0	PC3	ARP		0.001	N	3
	Successful	Hub0	PC4	ARP		0.001	N	4
	Failed	PC1	---	ICMP		0.002	N	5
	Successful	PC0	Hub0	ICMP		0.003	N	6

The hub repeats every frame on all ports. PC1, PC2 and PC4 receive the frame and drop it because it is not meant for them.

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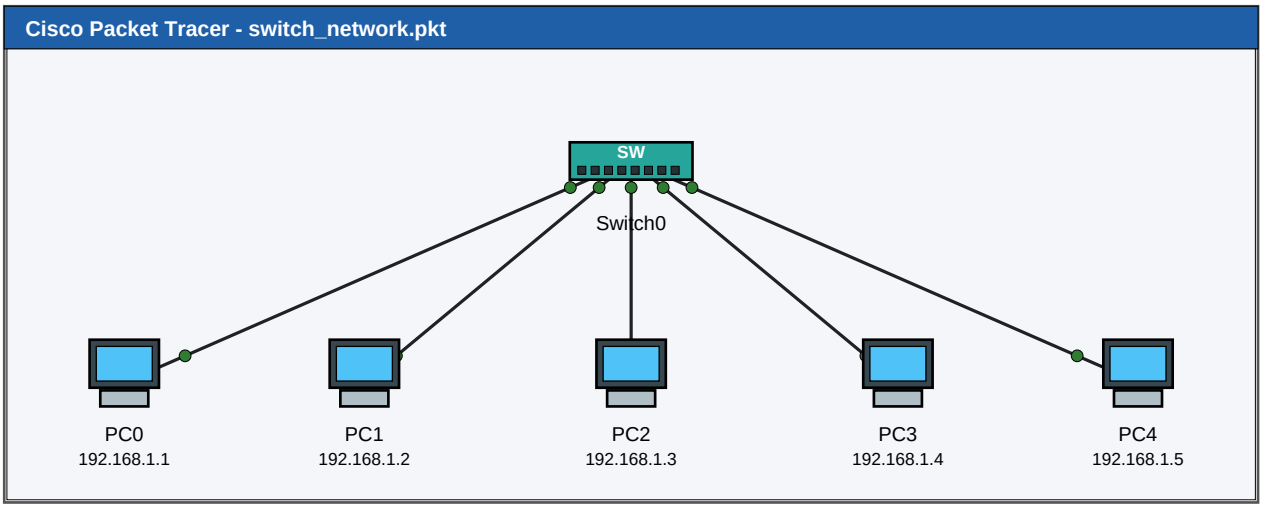


Experiment 2 : switch network in Packet Tracer

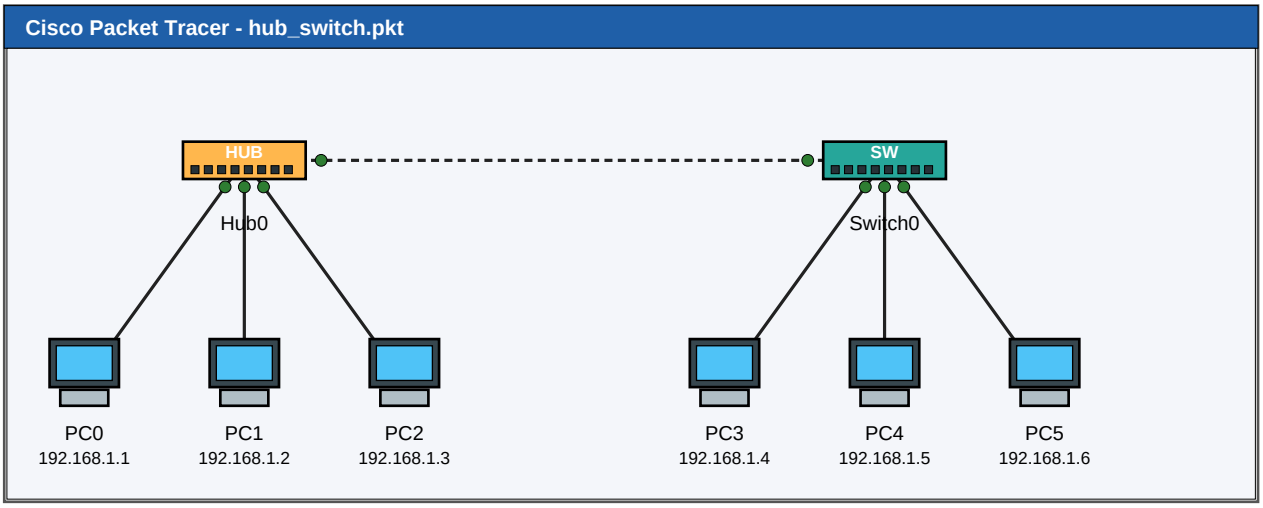
Switch0 - CLI

```
Switch>enable
Switch#show mac-address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----
   1    0001.4382.a3c1    DYNAMIC     Fa0/1
   1    0002.1779.b4d2    DYNAMIC     Fa0/4
   1    0060.2fa1.0c55    DYNAMIC     Fa0/2
Switch#
```

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Switch0	ARP		0.000	N	0
	Successful	Switch0	PC1	ARP		0.001	N	1
	Successful	Switch0	PC2	ARP		0.001	N	2
	Successful	Switch0	PC3	ARP		0.001	N	3
	Successful	PC3	Switch0	ARP		0.002	N	4
	Successful	PC0	Switch0	ICMP		0.003	N	5
	Successful	Switch0	PC3	ICMP		0.004	N	6

Only the first ARP broadcast is flooded. After the switch has learnt the MAC addresses, the ICMP frame goes only to the port of PC3.



Experiment 3 : hub and switch in the same network

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Hub0	ICMP		0.000	N	0

	Successful	Hub0	PC1	ICMP		0.001	N	1
	Successful	Hub0	PC2	ICMP		0.001	N	2
	Successful	Hub0	Switch0	ICMP		0.001	N	3
	Successful	Switch0	PC3	ICMP		0.002	N	4
	Successful	PC3	Switch0	ICMP		0.003	N	5
	Successful	Switch0	Hub0	ICMP		0.004	N	6

Point	**Hub**	**Switch**
OSI layer	Physical layer (Layer 1)	Data link layer (Layer 2)
Forwarding	Repeats the signal to all ports	Forwards the frame only to the destination port
Address table	None	MAC address table (learning)
Collision domain	One for all the ports	One for each port
Bandwidth	Shared among all hosts	Dedicated to each port
Duplex	Half duplex	Full duplex
Security / traffic	Anyone can capture the traffic	Traffic is limited to the two hosts

RESULT:

Thus the Cisco Packet Tracer environment was studied. Networks using only a hub, only a switch, and both a hub and a switch were designed and the packet flow was traced in simulation mode. The hub broadcasts every frame to all ports whereas the switch learns MAC addresses and forwards the frame only to the destination port.